

Dengyuhan Dai

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Profile

Fourth-year Applied Mathematics student at the University of California, Los Angeles. Academic interests include optimization, machine learning, and LLM algorithms, with experience in numerical methods and computational modeling.

Education

Imperial College London
MSc in Artificial Intelligence

Expected Sep 2026 -

University of California, Los Angeles
B.S. in Applied Mathematics

Sep 2024 – Present

- Overall GPA: 3.87/4.0
- **Selected Coursework:** Algorithms, Optimization, Numerical Methods, Advanced Linear Algebra, Probability, C++ Programming, Mathematical Modeling, Complex Analysis, Mathematical Analysis
- Transferred from UCSB in 2024 based on academic performance.

Research & Technical Experience

Traveling Salesman Point Art Math+Code+Art Project
Advised by Prof. Jiayin Lu

Dec 2025 – Present

- Developed an end-to-end pipeline converting continuous images into large-scale Euclidean Traveling Salesman Problem (TSP) instances through density field construction and adaptive point sampling based on LKH-3 solver.
- Designed multi-layer sampling strategies and incorporated image segmentation with SAM3 to enable divide-and-conquer preprocessing for high-resolution inputs.
- Applied dithering and structural constraints to balance geometric fidelity and computational tractability.

Independent Study in Topology Optimization & Scientific Machine Learning

Sep 2025 – Mar 2026

- Studying level-set and SIMP-based topology optimization methods with PDE constraints
- Implementing FEM-based optimization routines and sensitivity analysis
- Experimenting with FreeFEM++ for structural optimization
- Exploring surrogate modeling and neural-network acceleration for computational design

Guided Reading in Stochastic Processes
Advised by Prof. Vadim Markel

Aug 2023 – Dec 2023

- Studied fundamental stochastic process models and algorithms, including Markov chains and Poisson processes
- Explored applications of stochastic processes in statistical modeling and inference

Projects

Population Dynamics Modeling and Data Validation
Advised by Prof. Edith Zhang

Feb 2026 – Apr 2026

- Conducted qualitative and stability analysis of classical Lotka–Volterra systems and logistic-based extensions.
- Performed phase plane analysis, equilibrium classification, and long-term behavior investigation.
- Fitted models to ecological time-series data using numerical integration and parameter estimation techniques.
- Evaluated structural discrepancies between classical models and empirical data; proposed extensions including Allee effect and seasonal changing based on classical models.

Movie_Estimator: Bayesian Viewing Volume Inference System

Oct 2025 – Dec 2025

- Designed a probabilistic estimation framework combining weighted sampling and Beta–Binomial Bayesian correction.

- Modeled category-wise viewing distributions and aggregated posterior estimates.
- Integrated statistical inference with high-level semantic reasoning modules for adaptive correction.

Publications

Several Factors Affecting the Linear Regression Model

Nov 2023

Dengyuan Dai

10.1145/3653724.3653734 *Undergraduate exploratory work.*

Skills

Programming Languages: Python, C++, Java, HTML, Lua, $\text{\LaTeX}$, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, SciPy, FreeFEM++

Languages: Chinese (Native), English (Fluent), Japanese (JLPT N2)

Honors & Additional Activities

- Dean's Honors List(UCLA, UCSB)
- Independent Game Mod Developer (Lua / C++;)
- Selected member, Japan Academic Visit (2019)